

10) Creation of VHD on Windows and restoring on different host

Aim:

To create a VHD on windows and restore the same in different host

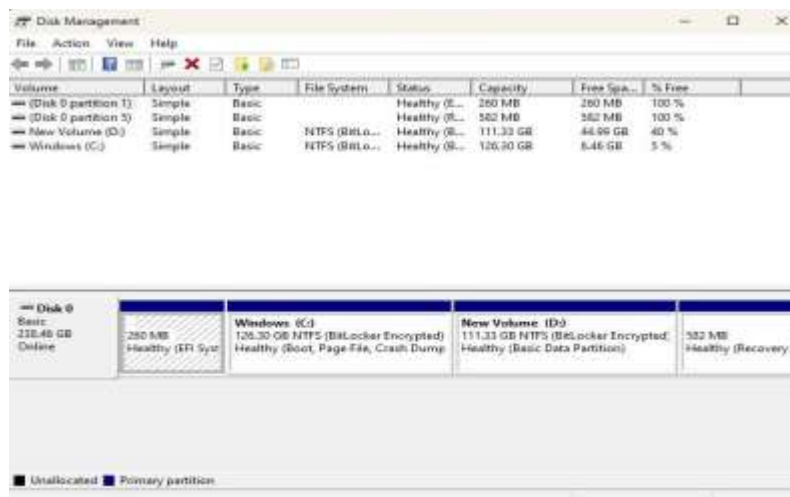
Steps:

Create a VHD on Windows

1. Press Win + R, type diskmgmt.msc, and press Enter

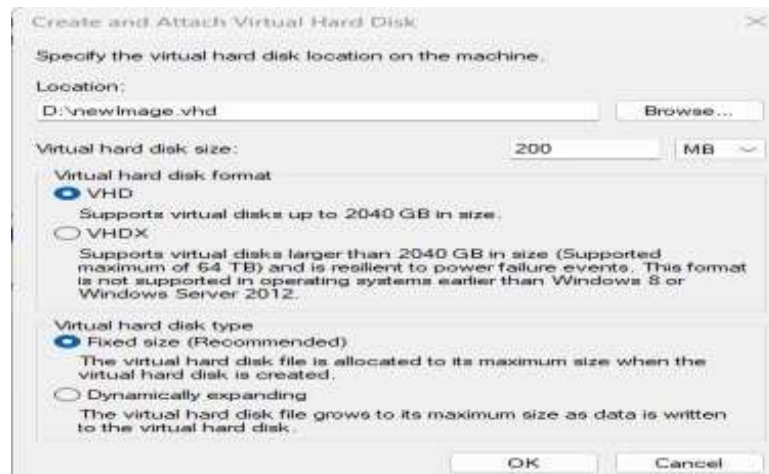


2. In Disk Management, go to the Action menu → Create VHD



3. Location → Browse and select where to save the .vhd or .vhdx file.
4. Virtual Hard Disk Size (decide based on what you need).
5. VHD format:VHD (compatible with older systems, max 2TB).

6. Type: Fixed size → Pre-allocates the full size. Faster runtime, but uses disk space upfront.
7. Dynamically expanding → Grows as data is added. Saves space initially.
8. Click OK → The VHD will be created and appear in Disk Management as "Not Initialized".

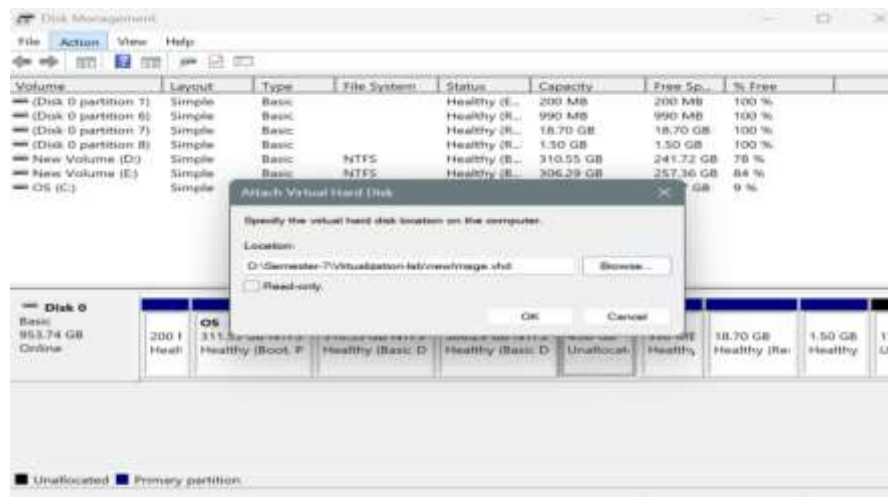


9. Right-click the new disk → Initialize Disk.
10. Create a new volume: Right-click unallocated space → New Simple Volume
→ Assign a drive letter → Format (NTFS recommended).
11. Now you can use it like a normal drive.

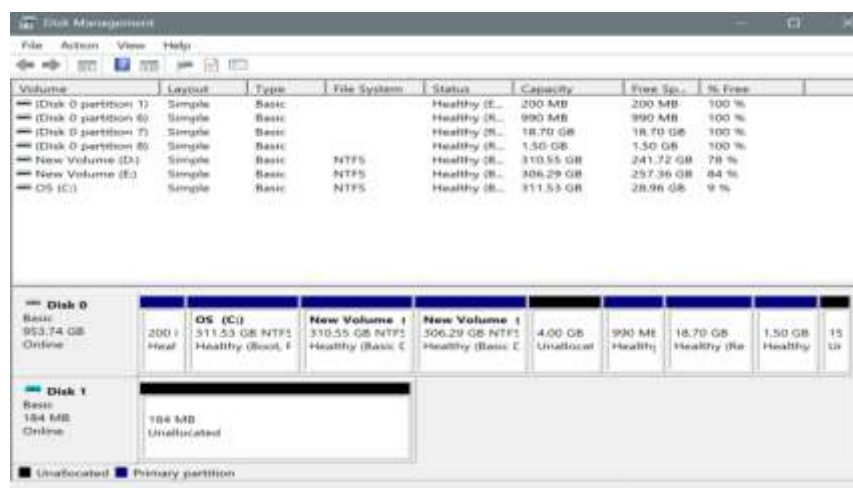


Steps to Use on Another Host

1. Copy your .vhd or .vhdx file to the new machine (via external HDD, network, or cloud).
2. On the new host: Open Disk Management → Action → Attach VHD.
3. Browse to the .vhd file and click OK.



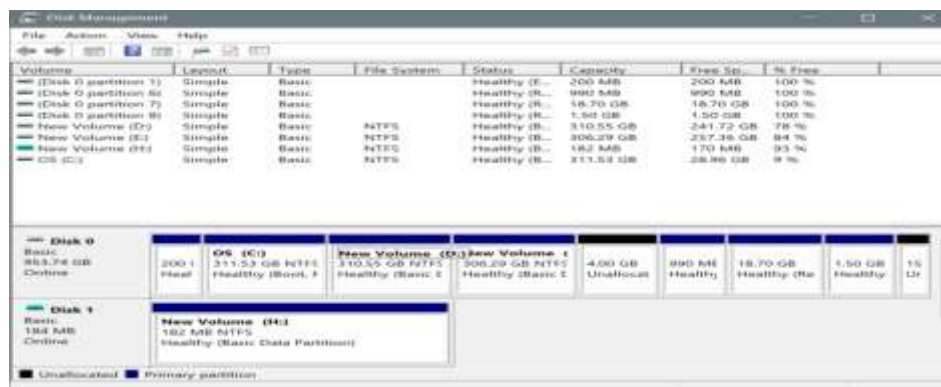
4. The VHD will appear as a new drive (with the same partitions/data).



5. If needed, assign a drive letter: Right-click the partition → Change Drive Letter and Paths → Assign a letter.



6. Now you can access files just like a normal disk.



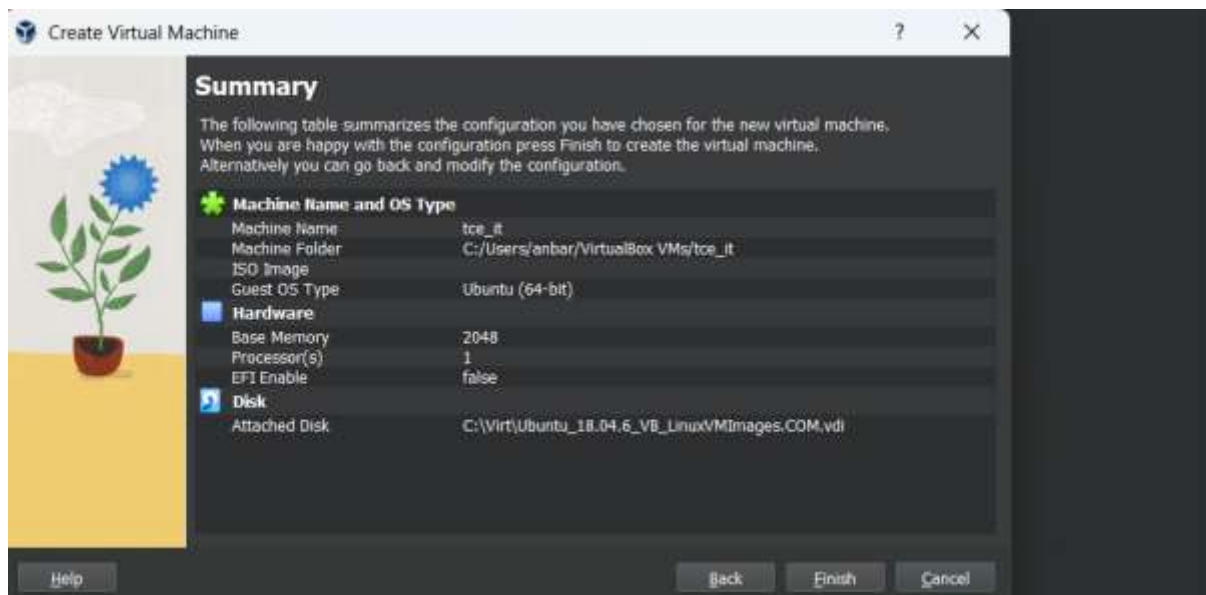
STORAGE VIRTUALIZATION:

Storage Virtualization

Step 1: Create a VM and Add Additional Storage

➤ Create a Linux VM

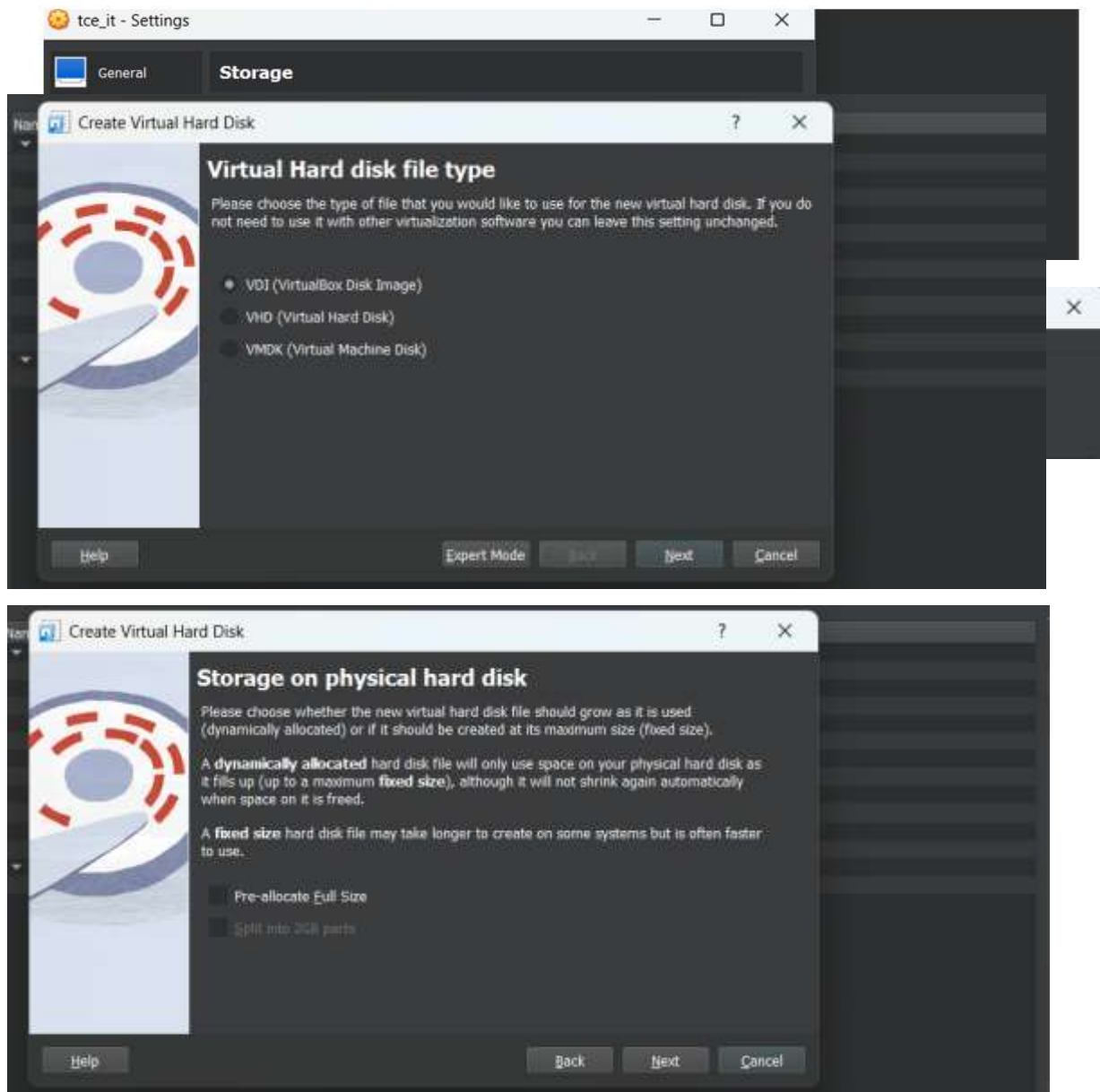
- Open VirtualBox → Click New → Enter VM Name, Type = Linux, Version = ubuntu.
- Allocate RAM (e.g., 2–4 GB).
- Hard Disk: Select “Use an existing virtual hard disk file”.
- Browse and select your VDI file.
- Click Finish.



➤ Add an Additional Virtual Disk

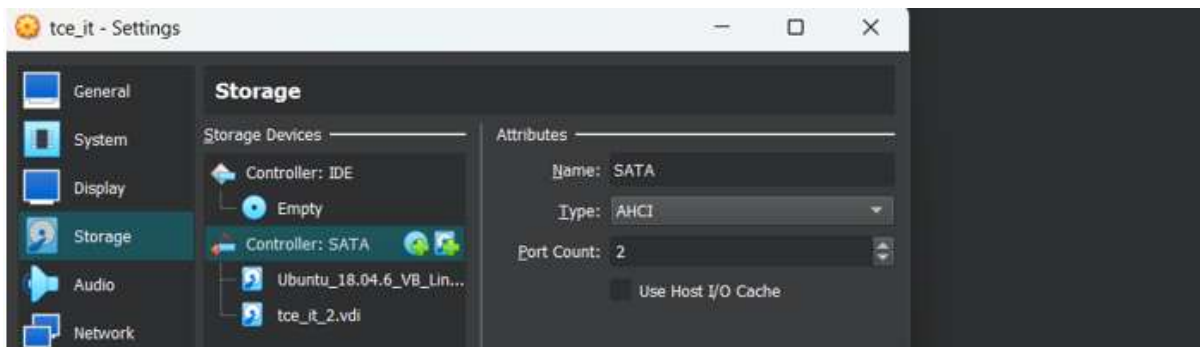
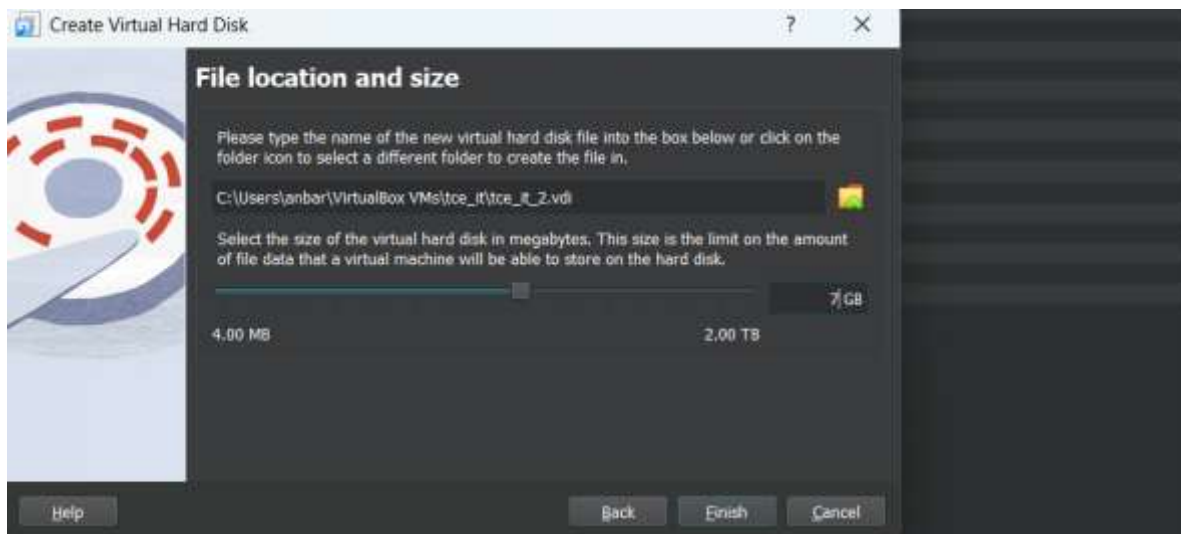
- This disk will be used for extending storage via LVM.
- In VirtualBox:
 1. Select the VM → Settings → Storage.

2. Click the Controller: SATA → Add Hard Disk → Create New Disk.



3. Choose VDI (VirtualBox Disk Image), Dynamically allocated, size = 7 GB.

4. Click Finish.



Step 2: Start the VM and Login as Root

```
ubuntu@ubuntu1804:~$ sudo -s
```

Step 3: Check Existing Disks and LVM

List all disks

fdisk -l

```
root@ubuntu1804:~# fdisk -l
Disk /dev/loop0: 2.5 MiB, 2621440 bytes, 5120 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes

Disk /dev/loop1: 65.1 MiB, 68259840 bytes, 133320 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes

Disk /dev/loop2: 61.8 MiB, 64770048 bytes, 126504 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
```



```

Disk /dev/sda: 512 GiB, 549755813888 bytes, 1073741824 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x67451bf8

```

Device	Boot	Start	End	Sectors	Size	Id	Type
/dev/sda1	*	2048	1073739775	1073737728	512G	8e	Linux LVM

```

Disk /dev/sdb: 7 GiB, 7516192768 bytes, 14680064 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes

```

List physical volumes

pvdisk

```

root@ubuntu1804:~# pvdisk
--- Physical volume ---
PV Name                /dev/sda1
VG Name                ubuntu-vg
PV Size                <512.00 GiB / not usable 2.00 MiB
Allocatable            yes (but full)
PE Size                4.00 MiB
Total PE               131071
Free PE                0
Allocated PE           131071
PV UUID                3F4Ims-mipI-J39Z-Lt6e-d2dv-a2Fz-wUzCld

```

List volume groups

vgdisk

```

root@ubuntu1804:~# vgdisk
--- Volume group ---
VG Name                ubuntu-vg
System ID
Format                lvm2
Metadata Areas         1
Metadata Sequence No   3
VG Access               read/write
VG Status               resizable
MAX LV                 0
Cur LV                 2
Open LV                 2
Max PV                 0
Cur PV                 1
Act PV                 1
VG Size                <512.00 GiB
PE Size                4.00 MiB
Total PE               131071
Alloc PE / Size         131071 / <512.00 GiB
Free PE / Size          0 / 0
VG UUID                3Mvbsr-MdcR-wlWG-r4GS-nouT-AMmM-33sdPM

```


List logical volumes

lvdisplay

```
root@ubuntu1804:~# lvdisplay
--- Logical volume ---
LV Path                /dev/ubuntu-vg/root
LV Name                 root
VG Name                 ubuntu-vg
LV UUID                 gDvWb8-E4oV-ogjV-UDT7-wx2V-gbCk-Z0FRjv
LV Write Access         read/write
LV Creation host, time  ubuntu, 2022-03-07 23:20:07 -0500
LV Status                available
# open                  1
LV Size                 <511.04 GiB
Current LE              130826
Segments                1
Allocation               inherit
Read ahead sectors      auto
- currently set to     256
Block device            253:0
```

```
--- Logical volume ---
LV Path                /dev/ubuntu-vg/swap_1
LV Name                 swap_1
VG Name                 ubuntu-vg
LV UUID                 PL8auK-uz9c-Wzlf-mCYt-Xcnl-SLki-7qcjWu
LV Write Access         read/write
LV Creation host, time  ubuntu, 2022-03-07 23:20:07 -0500
LV Status                available
# open                  2
LV Size                 980.00 MiB
Current LE              245
Segments                1
Allocation               inherit
Read ahead sectors      auto
- currently set to     256
Block device            253:1
```

Step 4: Create Physical Volume

pvcreeate /dev/sdb

```
root@ubuntu1804:~# pvcreeate /dev/sdb
Physical volume "/dev/sdb" successfully created.
```

Initializes the new disk for LVM.

Step 5: Extend the Volume Group

vgextend ubuntu-vg /dev/sdb

```
root@ubuntu1804:~# vgextend ubuntu-vg /dev/sdb
Volume group "ubuntu-vg" successfully extended
```

Replace cloud-vg with your existing Volume Group name.

This adds the new PV to your VG.

Step 6: Extend the Logical Volume (Root)

`lvresize --size +3G /dev/ubuntu-vg/root`

- +3G → amount of space to add
- /dev/cloud-vg/root → your logical volume

```
root@ubuntu1804:~# lvresize --size +3G /dev/ubuntu-vg/root
Size of logical volume ubuntu-vg/root changed from <511.04 GiB (130826 extents) to <514.04 GiB
(131594 extents).
Logical volume ubuntu-vg/root successfully resized.
```

Alternative:

`lvextend -L +3G /dev/ubuntu-vg/root`

Step 7: Check Current Disk Usage

`df -h`

Shows current available space on /.

```
root@ubuntu1804:~# df -h
Filesystem      Size  Used Avail Use% Mounted on
udev            969M   0  969M   0% /dev
tmpfs           199M  1.6M  198M   1% /run
/dev/mapper/ubuntu--vg-root 503G  5.9G  471G   2% /
tmpfs           994M   0  994M   0% /dev/shm
tmpfs           5.0M  4.0K  5.0M   1% /run/lock
tmpfs           994M   0  994M   0% /sys/fs/cgroup
/dev/loop0       2.5M  2.5M   0 100% /snap/gnome-system-monitor/163
/dev/loop3       768K  768K   0 100% /snap/gnome-characters/741
/dev/loop1        66M   66M   0 100% /snap/gtk-common-themes/1515
```

```
/dev/loop2        62M   62M   0 100% /snap/core20/1081
/dev/loop4       219M  219M   0 100% /snap/gnome-3-34-1804/77
/dev/loop6       2.5M  2.5M   0 100% /snap/gnome-calculator/884
/dev/loop7       640K  640K   0 100% /snap/gnome-logs/106
/dev/loop8        56M   56M   0 100% /snap/core18/2284
/dev/loop9       242M  242M   0 100% /snap/gnome-3-38-2004/70
/dev/loop10      219M  219M   0 100% /snap/gnome-3-34-1804/72
/dev/loop12      128K  128K   0 100% /snap/bare/5
/dev/loop11      249M  249M   0 100% /snap/gnome-3-38-2004/99
/dev/loop13      2.7M  2.7M   0 100% /snap/gnome-system-monitor/174
/dev/loop14      2.7M  2.7M   0 100% /snap/gnome-calculator/920
/dev/loop15       44M   44M   0 100% /snap/snapd/14978
/dev/loop16       66M   66M   0 100% /snap/gtk-common-themes/1519
/dev/loop17       62M   62M   0 100% /snap/core20/1361
/dev/loop18      768K  768K   0 100% /snap/gnome-characters/726
tmpfs           199M   28K  199M   1% /run/user/121
tmpfs           199M   40K  199M   1% /run/user/1000
/dev/loop19       56M   56M   0 100% /snap/core18/2952
/dev/loop20       51M   51M   0 100% /snap/snapd/25202
```

Step 8: Resize the Filesystem

resize2fs /dev/mapper/cloud--vg-root

```
root@ubuntu1804:~# resize2fs /dev/mapper/ubuntu--vg-root
resize2fs 1.44.1 (24-Mar-2018)
Filesystem at /dev/mapper/ubuntu--vg-root is mounted on /; on-line resizing required
old_desc_blocks = 64, new_desc_blocks = 65
The filesystem on /dev/mapper/ubuntu--vg-root is now 134752256 (4k) blocks long.
```

Resizes the filesystem to use the newly added space.

Step 9: Verify Disk Space

df -h

```
root@ubuntu1804:~# df -h
Filesystem      Size  Used Avail Use% Mounted on
udev            969M   0    969M   0% /dev
tmpfs           199M  1.6M  198M   1% /run
/dev/mapper/ubuntu--vg-root 505G  5.9G  474G   2% /
tmpfs           994M   0    994M   0% /dev/shm
tmpfs           5.0M  4.0K   5.0M   1% /run/lock
tmpfs           994M   0    994M   0% /sys/fs/cgroup
/dev/loop0      2.5M  2.5M   0 100% /snap/gnome-system-monitor/163
/dev/loop3      768K  768K   0 100% /snap/gnome-characters/741
/dev/loop1       66M   66M   0 100% /snap/gtk-common-themes/1515
/dev/loop2       62M   62M   0 100% /snap/core20/1081
/dev/loop4      219M  219M   0 100% /snap/gnome-3-34-1804/77
/dev/loop6      2.5M  2.5M   0 100% /snap/gnome-calculator/884
/dev/loop7      640K  640K   0 100% /snap/gnome-logs/106
/dev/loop8       56M   56M   0 100% /snap/core18/2284
/dev/loop9      242M  242M   0 100% /snap/gnome-3-38-2004/70
/dev/loop10     219M  219M   0 100% /snap/gnome-3-34-1804/72
/dev/loop12     128K  128K   0 100% /snap/bare/5
/dev/loop11     249M  249M   0 100% /snap/gnome-3-38-2004/99
/dev/loop13     2.7M  2.7M   0 100% /snap/gnome-system-monitor/174
/dev/loop14     2.7M  2.7M   0 100% /snap/gnome-calculator/920
/dev/loop15      44M   44M   0 100% /snap/snapd/14978
/dev/loop16      66M   66M   0 100% /snap/gtk-common-themes/1519
/dev/loop17      62M   62M   0 100% /snap/core20/1361
/dev/loop18     768K  768K   0 100% /snap/gnome-characters/726
tmpfs           199M  28K   199M   1% /run/user/121
tmpfs           199M  40K   199M   1% /run/user/1000
/dev/loop19      56M   56M   0 100% /snap/core18/2952
/dev/loop20      51M   51M   0 100% /snap/snapd/25202
root@ubuntu1804:~#
```

Confirm that the root filesystem now has additional 3 GB.

Result:

Thus we have created VHD on windows and restored it on another machine successfully.